Akshit Achara

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PhD researcher studying how vision models represent data and fail. My work spans shortcut learning, representation learning, and robust generalisation across 2D and 3D settings, with experience building deployed AI systems in industrial environments.

Experience

King's College London London, UK
PhD Researcher *Oct 2024 – Present*

- Investigate how learned models form, align, and misuse predictive representations, with emphasis on failure modes that remain hidden under standard aggregate metrics.
- Develop diagnostics for spurious-signal reliance across representations, attribution maps, and dense prediction outputs, including shortcut localisation and semantic label-flip detection.
- Apply these ideas across vision, medical imaging, and language settings, spanning representation alignment, segmentation robustness and demographic predictability.

GE Research / GE Healthcare Bangalore, India
Research Engineer, Applied AI Research *Oct 2020 – Jul 2024*

- Developed vision systems for structural defect detection in field equipment, including prediction of near-invisible cracks that were difficult to identify manually.
- Partnered with oncologists to develop annotation guidelines, negation handling, and domain ontologies for longitudinal oncology records, and built patient-level representations to support downstream tasks including summarisation, family history extraction, entity linking, structured information extraction, and clinical trial matching across the patient journey.
- Led development of an end-to-end clinical trial matching system spanning trial ingestion, eligibility structuring, patient entity linking, and ranking, with emphasis on error analysis and robustness of matching decisions.
- Represented GE Healthcare at G20 events in India, presenting applied AI initiatives to cross-stakeholder audiences.

Education

King's College London London, UK
PhD, Biomedical Engineering and Imaging Sciences *Oct 2024 – Present*

Birla Institute of Technology and Science (BITS Pilani) Pilani, India
M.Sc. Mathematics *Aug 2016 – Jul 2020*

Selected Publications

†Joint first authors. ‡Joint supervision.

- **Achara, A.**, Gaintseva, T., Mahaut, M., Chakraborty, P., Johansson, V. S., Barsbey, M., Rodolà, E., and Crisostomi, D. *Multi-Way Representation Alignment*. ICML (2026), ReAlign @ ICLR (2026).
- Avci, M. Y.†, **Achara, A.**†, King, A.‡, Cardoso, J.‡, for the Alzheimer's Disease Neuroimaging Initiative. *Understanding Sources of Demographic Predictability in Brain MRI via Disentangling Anatomy and Contrast*. arXiv (2026). Submitted to MICCAI (2026).
- **Achara, A.**, Yathathugoda, Y., Byrne, N., Antonelli, M., Puyol Anton, E., Hammers, A., and King, A. P. *Right Regions, Wrong Labels: Semantic Label Flips in Segmentation under Correlation Shift*. CAO @ ICLR (2026).
- **Achara, A.**, Triantafillou, P., Puyol-Antón, E., Hammers, A., King, A. P., for the Alzheimer's Disease Neuroimaging Initiative. *Localising Shortcut Learning in Pixel Space via Ordinal Scoring Correlations for Attribution Representations (OSCAR)*. arXiv (2025).

- **Achara, A.**, Puyol Anton, E., Hammers, A., and King, A. P. *Invisible Attributes, Visible Biases: Exploring Demographic Shortcuts in MRI-based Alzheimer's Disease Classification*. FAIMI @ MICCAI (2025).
- **Achara, A.**, and Chhabra, A. *Watching the AI Watchdogs: A Fairness and Robustness Analysis of AI Safety Moderation Classifiers*. NAACL (2025).
- **Achara, A.**, Sasidharan, S., and N, G. *Efficient Biomedical Entity Linking: Clinical Text Standardisation with Low-Resource Techniques*. BioNLP @ ACL (2024).

Patents (US Applications)

- **Achara, A.**, Goravar, S., Sasidharan, S., and Kanamarlapudi, A. *Clinical context centric natural language processing solutions*. Allowed US Patent App. 18/183,295 (2024).
- Sasidharan, S., **Achara, A.**, Kanamarlapudi, A., Goravar, S., and Bhoi, K. *Systems and methods for intelligent medical editors*. US Patent App. 18/906,528 (2025).
- Ghose, S., Sasidharan, S., Cho, S., **Achara, A.**, Mullick, R., Kanamarlapudi, A., Ginty, F., Bhardwaj, A. R., Mani, S., and Sarachan, B. *Natural language based clinical recommendation system*. US Patent App. 18/468,593 (2025).

Awards & Service

- **UKRI-funded DRIVE-Health CDT Scholarship**, King's College London, Oct 2024– Sept 2028.
- Best Team Award, LOGML 2025.
- Top Reviewer, NeurIPS 2025.
- Reviewer: NeurIPS 2024, ICML 2025, AISTATS 2026.

Skills

- **Programming:** Python, C/C++, R, Java
- **Frameworks:** PyTorch, Lightning, Transformers
- **Research Areas:** computer vision, representation learning, shortcut learning, interpretability